

# Effects of Fatty Acids and Hormones on Fatty Acid Metabolism and Gluconeogenesis in Bovine Hepatocytes

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## ABSTRACT

Primary cultures of hepatocytes were used to study the effects of extracellular oleate concentration and hormones on fatty acid metabolism and gluconeogenesis. Rates of oleate uptake and oxidation to acid-soluble products varied linearly as oleate concentrations increased (0.1 to 2 mM), but rates of triglyceride accumulation varied quadratically. Insulin increased the proportion of oleate that was esterified by 22% without affecting the formation of acid-soluble products. Cells incubated with 2 mM [1-<sup>14</sup>C]oleate for 24 h eliminated 9.6% of the labeled intracellular lipid as acid-soluble products in the following 24 h when no oleate was present during depletion and eliminated 7.7% when 2 mM oleate was present. Insulin reduced labeled triglyceride depletion by 49%. Gluconeogenesis from [2-<sup>14</sup>C] propionate was depressed by 24%, and formation of acid-soluble products was increased by 46% in cells infiltrated with lipid because of previous exposure to 2 mM oleate for 45 h. Rates of gluconeogenesis from propionate were reduced 23% when 2 mM oleate was present during the 3-h period that gluconeogenesis was measured, and the effect was not modified by lipid infiltration. Lipid infiltration influenced hepatic function, and insulin regulated hepatic triglyceride concentration.

(**Key words:** hepatic ketogenesis, hepatic fatty acid oxidation, hepatic triglyceride mobilization, fatty liver)

**Abbreviation key:** ASP = acid-soluble product, E<sub>3</sub> = estriol, ICL = intracellular lipids, TG = triglyceride.

## INTRODUCTION

Fatty liver is usually associated with a pronounced negative energy balance at the beginning of lactation

in high yielding dairy cows. Recent studies (4, 40) from our laboratory suggest that some fat infiltration occurs prior to and during calving. This infiltration may result from elevated NEFA caused by endocrine changes that are associated with calving and by reduced feed intake during the final weeks preceding parturition. Bertics et al. (4) found that force-feeding dairy cows during the last 2 wk of gestation reduced fat and triglyceride (TG) content of the liver at calving, which supports the idea that fatty acid mobilization at the end of gestation is a determinant. As parturition approaches, plasma concentrations of NEFA increase from <0.5 to 2 mM (4). It is not known whether the hepatic uptake of fatty acids and their relative partition between oxidation and esterification within the liver is constant or changes over the physiological range of NEFA concentrations.

Changes in plasma insulin, growth hormone, prolactin, estrogens, or progesterone (10, 11, 33) at the end of gestation may influence the metabolic fate of fatty acids within the liver. In vitro data in non-ruminants have indicated that estrogens increase the incorporation of media fatty acids into hepatic TG (12), and in vivo data for ruminants have shown that estrogens accelerate fat infiltration of the liver during starvation (21, 28). Insulin deficiency enhanced ketogenesis in monogastrics and ruminants (23), and exogenous insulin promoted TG synthesis (15). Glucagon, when administered in vivo to humans, tended to increase plasma ketones, NEFA, and glycerol concentrations (8). However, whether these effects are via direct actions on liver, on adipose tissue, or both is not known.

Most studies of lipid metabolism in ruminant hepatocytes have been conducted in short-term, suspended hepatocyte cultures (2, 26). Monolayer hepatocyte cultures were only recently made available for research on ruminants (13, 14, 18), and those cultures provide a more suitable system for the study of the direct regulation of fatty acid utilization and the consequences of prolonged exposure to high extracellular concentrations of fatty acids. Plated cells accumulated TG after incubation in media that had been supplemented with fatty acids (37) and could be

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used to study the consequences of lipid engorgement on cell function and the metabolic fate of stored TG.

The objectives of the present study were 1) to test whether hepatic fractional uptake and partitioning of fatty acids toward oxidation or esterification remains constant over the physiological range of plasma NEFA concentrations; 2) to determine the role of insulin, glucagon, and estrogens in the regulation of fatty acid uptake and metabolism by bovine hepatocytes; 3) to estimate the *in vitro* rates of disposal of hepatic intracellular TG and its regulation; and 4) to examine how TG infiltration induced by NEFA might affect gluconeogenesis and fatty acid metabolism in bovine hepatocytes.

## MATERIALS AND METHODS

### Experiment 1: Oleate Uptake and Partition

Bovine hepatocytes were isolated from the caudate process of three 8- to 10-d-old Holstein male calves and seeded onto 30-mM Falcon Primaria culture dishes (Becton Dickinson, Lincoln Park, NJ) using the procedure described previously (13). Four hours after seeding, the initial plating media were removed and replaced with 1.5 ml of fresh, basal culture medium [arginine-free, Dulbecco's modified Eagle's medium containing 0.5 mM ornithine, 24 mM sodium bicarbonate, 10 mM HEPES, 4.0 mM glutamine, 1.0 mM pyruvate, 5.5 mM glucose, 100 IU of penicillin/ml, 100  $\mu$ g of streptomycin/ml, 250 ng of amphotericin/ml, 3% BSA (Cohn fraction V and essentially free of fatty acids), 100 nM dexamethasone, 2 mM dl-carnitine, and supplemental [ $1\text{-}^{14}\text{C}$ ]oleate (0, 0.1, 0.5, 1, or 2 mM)]. Effects of hormones were studied using hepatocytes incubated with 1 and 2 mM oleate by adding estriol ( $\text{E}_3$ ; 1 or 10 nM), insulin (10 nM), glucagon (1 nM), or no hormonal addition. Estriol was chosen among other estrogens available because of its relatively slow metabolism by hepatocytes. A stock of 10 mM  $\text{E}_3$  in ethanol was prepared and subsequently diluted in saline to obtain a 0.75  $\mu$ M  $\text{E}_3$  solution. The final concentration of ethanol in the culture medium was <20 nM. Insulin, glucagon, dexamethasone, and all other reagents were as described previously (13).

The incubations were stopped at 4, 24, or 48 h by removing the media from the plates and washing them with 1 ml of cold Krebs-Ringer bicarbonate solution. Media from plates that were cultured for 48 h were collected and replaced by fresh media at 24 h. The 24- and 48-h media and washing buffer from each plate were frozen at  $-18^\circ\text{C}$  until assay for label incor-

poration into acid-soluble products (ASP) as described by Kleppe et al. (26). The plates were then covered with 1 ml of ice-cold dissociation buffer (2.68 mM KCl, 1.47 mM  $\text{KH}_2\text{PO}_4$ , 137 mM NaCl, 8.06 mM  $\text{Na}_2\text{HPO}_4$ , and 1.0 mM EDTA; pH 7.4) and frozen. Cells were later thawed, scraped from the plates, homogenized by sonication, and analyzed for DNA content (29). Neutral lipids were extracted (19) and brought to a final volume of 100  $\mu$ l in chloroform:methanol (3:1); an aliquot (20  $\mu$ l) was loaded onto polyester-backed, Silica Gel 60, thin-layer chromatography plates (POLYGRAM SILG 805013; Macherey-Nagel, Dueren, Germany) and developed with a petroleum ether:diethyl ether:acetic acid (80:20:1) mobile phase. The chromatographic plates were stained with rhodamine G, and the fragments containing the TG fractions were cut, placed into scintillation vials filled with Poly-Fluor scintillation cocktail (Packard Instrument Co., Downers Grove, IL), and counted (Tri-carb 4000 series liquid scintillation spectrometer; Packard Instrument Co.) to measure oleate incorporation into cellular TG. An aliquot of the Folch extract was also counted to measure oleate incorporation into total intracellular lipids (ICL). The total uptake of oleate was calculated as the sum of oleate converted to ASP and ICL. This culture system did not permit measurement of oleate oxidation to  $\text{CO}_2$ ; previous studies (2) using short-term suspensions of hepatocytes indicated that  $\text{CO}_2$  formation accounted for 1 to 3% of oleate metabolism.

Oleate uptake and oleate conversion to ASP, intracellular TG, and ICL (nanomoles per microgram of DNA per hour) were determined for each treatment within cell preparation (calf) using the slope of the regression line of oleate transformed (nanomoles per microgram of DNA per hour) on time (hour). The proportion of oleate partitioned between ASP and TG was calculated by dividing the respective rates of formation of ASP and TG by the rate of oleate uptake. The ratio of the rates of ASP formation to TG esterification reflected the relative partition of fatty acids between oxidation and esterification.

### Experiment 2: Disposal of ICL

The rate of intracellular TG disappearance was measured in cells with a previously labeled pool of intracellular TG. These cells were incubated for the initial 24 h (TG loading period) in the presence of 2 mM [ $1\text{-}^{14}\text{C}$ ]oleate and either 0 or 10 nM  $\text{E}_3$ . Media were removed; fresh media containing 0 or 2 mM oleate without hormones or supplemented with  $\text{E}_3$  (1 or 10 nM), insulin (10 nM), or glucagon (1 nM) were added; and incubations were continued for an addi-

tional 24 h (from 24 to 48 h) to determine  $^{14}\text{C}$ [TG] depletion. Therefore, the two values of  $\text{E}_3$  applied during TG loading were factorialized with two values of oleate and five different hormonal additions during depletion. The amount of intracellular TG that was present at time zero of the depletion period was estimated from a set of eight plates per loading treatment that were terminated at the end of the loading period. The incubations were terminated, and label incorporation into cell TG and ICL and ASP in the media were determined as described previously. The total amount of label that was released into the media was determined by counting an aliquot of the media. The amount of intracellular label that was released because of cell lysis was estimated by multiplying the micrograms of DNA lost per plate over the depletion period times the amount of intracellular label per microgram of DNA at the beginning of the depletion period. For each plate, the ASP output and the lipids remaining intracellularly were expressed as a percentage of the intracellular label present per plate at the end of the loading period. Each treatment combination was tested on triplicate plates within cell preparations from two calves.

In order to check whether lipids had been exported from the cells as lipoprotein, depletion media were collected from plates representing one calf and were fractionated by gel filtration chromatography using a 92-cm column (K26/200; Pharmacia, Piscataway, NJ) containing Bio-Gel A5-M (200 to 2400 mesh; Bio-Rad Lab, Hercules, CA). Column buffer was eluted at 0.25 ml/min and collected in 6-ml fractions (22).

### Experiment 3: Ketogenesis and Gluconeogenesis in Cells Infiltrated with Lipid

The effects of intracellular TG accumulation on gluconeogenesis from propionate and on fatty acid uptake and metabolism were studied using two cell preparations as described previously. Four hours after plating, cells were incubated with basal media that had been supplemented with 0 or 2 mM oleate. Media were refreshed at 12, 24, and 36 h and were replaced at 45 h with basal media that had been supplemented with 2.5 mM  $[2\text{-}^{14}\text{C}]$ propionate in the presence of 0 or 2 mM oleate and 10 nM insulin, 1 nM glucagon, or no hormonal addition. At 48 h, incubations were stopped, media and cells were collected, and rates of propionate conversion to media glucose (nanomoles of propionate incorporated into glucose per microgram of DNA per hour) were measured as described previously (13). Neutral lipids from these cells were extracted and separated as described previ-

ously to estimate propionate conversion to intracellular TG glycerol. Parallel incubations cultured with 2 mM  $[1\text{-}^{14}\text{C}]$ oleate in the presence of 0 or 2.5 mM propionate and 10 nM insulin, 1 nM glucagon, or no hormonal addition were used to measure the formation of ASP.

### Statistics

For Experiment 1, when media contained different concentrations of oleate, sources of variation were calf (i.e., block), oleate concentration, time, and the interaction of oleate concentration and time. The rates of oleate uptake and conversion to ASP and TG were derived for each calf by treatment combination by linear regression on time. For each treatment combination (oleate concentration by hormonal addition), three independent estimates of each rate were obtained. The dependency of these rate estimates on oleate concentration was tested with a second-degree polynomial linear regression, taking into account the block effect. When cells were incubated with 1.0 or 2.0 mM oleate and various hormones, sources of variation were calf, oleate concentration, hormone, and the interaction of oleate and hormones.

For Experiment 2, sources of variation were calf, presence of  $\text{E}_3$  during the loading period, presence of oleate during depletion, hormone application during depletion, and interactions among in vitro treatments. Effects of hormone were tested by contrasts against the control (i.e., no hormones were added during the depletion period).

Rates of gluconeogenesis in Experiment 3 were analyzed using a model containing calf, preexposure to oleate, concurrent exposure to oleate, hormone treatment, and interactions among preexposure to oleate, concurrent exposure to oleate, and hormone treatments. Rates of formation of ASP were analyzed using a similar model, except concurrent exposure to oleate was replaced by concurrent exposure to propionate. For each experiment, all variables in the model were tested using the residual error mean square. Analysis of variance was conducted using the GLM procedure of SAS (36). Significant effects were declared when  $P < 0.05$ .

## RESULTS AND DISCUSSION

### Experiment 1: Oleate Uptake and Partition

**Effects of NEFA concentration.** At every sampling time, the mass of oleate oxidized to ASP or converted to ICL and TG increased as oleate concen-

TABLE 1. Oleate converted to media acid-soluble products (ASP), intracellular triglyceride (TG), and total intracellular lipids (ICL) (Experiment 1).<sup>1</sup>

| Parameter<br>and incubation<br>period <sup>2</sup> | Oleate in media                  |                   |           |                    |           |                    |           |                    |
|----------------------------------------------------|----------------------------------|-------------------|-----------|--------------------|-----------|--------------------|-----------|--------------------|
|                                                    | 0.1 mM                           |                   | 0.5 mM    |                    | 1.0 mM    |                    | 2.0 mM    |                    |
|                                                    | (nmol of oleate/ $\mu$ g of DNA) |                   |           |                    |           |                    |           |                    |
|                                                    | $\bar{X}$                        | SE                | $\bar{X}$ | SE                 | $\bar{X}$ | SE                 | $\bar{X}$ | SE                 |
| ASP                                                |                                  |                   |           |                    |           |                    |           |                    |
| 0 to 4 h                                           | 0.82                             | 0.46 <sup>c</sup> | 3.04      | 1.34 <sup>bc</sup> | 4.40      | 2.17 <sup>ab</sup> | 6.15      | 2.35 <sup>a</sup>  |
| 0 to 24 h                                          | 2.29                             | 1.00 <sup>c</sup> | 9.94      | 3.21 <sup>bc</sup> | 17.28     | 5.24 <sup>ab</sup> | 26.55     | 10.85 <sup>a</sup> |
| 24 to 48 h                                         | 2.29                             | 1.26 <sup>c</sup> | 13.54     | 5.54 <sup>bc</sup> | 18.72     | 6.07 <sup>ab</sup> | 31.48     | 11.78 <sup>a</sup> |
| Intracellular TG                                   |                                  |                   |           |                    |           |                    |           |                    |
| 0 to 4 h                                           | 0.22                             | 0.05 <sup>c</sup> | 0.99      | 0.35 <sup>c</sup>  | 2.31      | 0.27 <sup>b</sup>  | 8.29      | 0.76 <sup>a</sup>  |
| 0 to 24 h                                          | 0.61                             | 0.18 <sup>b</sup> | 2.11      | 0.31 <sup>b</sup>  | 5.04      | 0.64 <sup>b</sup>  | 19.32     | 4.26 <sup>a</sup>  |
| 0 to 48 h                                          | 1.43                             | 0.62 <sup>c</sup> | 5.75      | 2.37 <sup>bc</sup> | 12.96     | 2.90 <sup>b</sup>  | 49.30     | 9.76 <sup>a</sup>  |
| Total ICL                                          |                                  |                   |           |                    |           |                    |           |                    |
| 0 to 4 h                                           | 0.67                             | 0.28 <sup>c</sup> | 2.52      | 1.04 <sup>bc</sup> | 4.34      | 0.31 <sup>b</sup>  | 11.49     | 2.31 <sup>a</sup>  |
| 0 to 24 h                                          | 1.92                             | 0.79 <sup>b</sup> | 4.62      | 1.30 <sup>b</sup>  | 8.45      | 1.75 <sup>b</sup>  | 27.50     | 7.82 <sup>a</sup>  |
| 0 to 48 h                                          | 4.57                             | 2.13 <sup>c</sup> | 12.14     | 5.09 <sup>bc</sup> | 21.35     | 4.30 <sup>b</sup>  | 65.38     | 13.26 <sup>a</sup> |

a,b,c Means within the same row with different superscripts differ ( $P < 0.01$ ).

<sup>1</sup>Values are means of three observations. Plated calf hepatocytes were cultured in Dulbecco's modified Eagle's medium supplemented with four concentrations of [ $^{14}$ C]oleate for 4, 24, or 48 h.

<sup>2</sup>For cells incubated for 48 h, ASP was measured in media removed at 24 and 48 h. In contrast, cells were only harvested at 48 h; therefore, a 24- to 48-h measurement was not possible.

tration in the media increased (Table 1). The hourly rate of oleate uptake and conversion to ASP and TG within oleate concentration remained stable throughout the 48-h incubations;  $R^2$  of the regressions ranged from 0.9999 to 0.8150. These rates were then regressed on the oleate concentrations of the media to test whether fatty acid uptake and utilization were proportional to the available substrate. The rate of oleate uptake and formation of ASP increased linearly ( $P < 0.05$ ) as extracellular oleate concentrations increased; there was a positive quadratic relationship ( $P < 0.05$ ; Figure 1) between the concentration of extracellular oleate and the rate of TG accumulation. The linear uptake of oleate was also reported for liver slices (33). However, Payne and Masters (33) did not report significant differences in oleate metabolism at two oleate concentrations. During those brief incubations, over 50% of the fatty acids taken up by the cells were present as cytoplasmic FFA. In our experiment, the intracellular FFA accounted for an insignificant proportion of total intracellular label at each time and oleate concentration tested (data not shown). In vitro experiments with suspended goat hepatocytes (26) also showed no accumulation of intracellular FFA. The large portion of FFA taken up by liver slices (33) might have been due to the accumulation of FFA within the extracellular spaces of liver tissue, which might have masked any differences in partitioning of fatty acids caused by the oleate concentrations of the media.

The ratio between the rates of oleate conversion to ASP and TG changed considerably over the range of oleate concentrations used in the experiment; the maximum proportion was converted to ASP when the

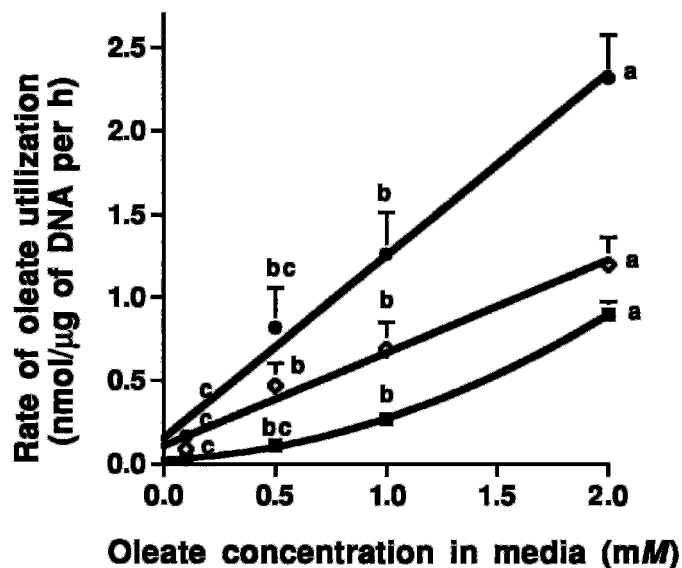


Figure 1. Rates of oleate uptake and conversion to acid-soluble products (ASP) and intracellular triglyceride (TG) in bovine hepatocytes. Means within a parameter (e.g., ASP) that do not share a common letter (a, b, or c) differ ( $P < 0.05$ ). ● = Uptake =  $0.517 + 1.078x$ ,  $r^2 = 0.888$ ; ◇ = ASP =  $0.297 + 0.583x$ ,  $r^2 = 0.877$ ; ■ = TG =  $0.059 + 0.042x + 0.197x^2$ ,  $r^2 = 0.911$ .

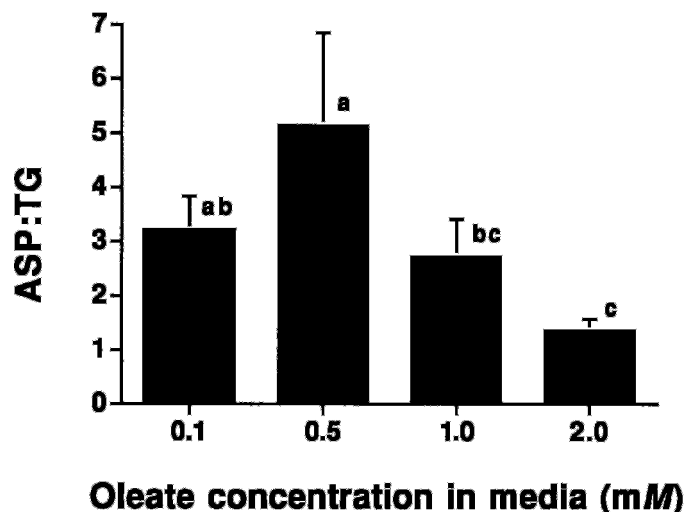


Figure 2. Ratio between oleate converted to acid-soluble products (ASP) and to intracellular triglyceride (TG) in bovine hepatocytes. Means that do not share a common letter (a, b, or c) differ ( $P < 0.05$ ).

extracellular concentrations of oleate were around 0.5 mM (Figure 2). Fatty acid uptake by the liver seemed to be dependent on concentration and was not regulated (17). A significant positive correlation between NEFA concentrations and liver lipid and TG content has been shown in vivo (4, 24, 35). Whether the rate of fatty acid oxidation becomes saturated while esterification continues at high rates of fatty acid uptake is unknown (43). Our findings suggest that, as plasma NEFA increased toward the upper range of physiological concentrations, the liver tended to increase its rate of TG accumulation without limiting its output of ASP. Conversely, Armentano et al. (2) found that increasing the rate of ASP formation did not limit fatty acid esterification.

The acceleration in rates of net TG accumulation observed at high oleate concentrations could have resulted from a decrease in the rate of lipolysis while the rates of esterification remained unchanged. Lipolysis in bovine adipose tissue was inhibited in vitro by  $\beta$ -hydroxybutyrate concentrations between 2 and 4 mM (31). In Experiment 1, the concentration of ASP in the media after 20 to 24 h of incubation in 2 mM oleate was  $>2$  mM. However, increased TG accumulation caused by decreased lipolysis is unlikely because rates of lipolysis of intracellular TG in bovine hepatocytes were too low (Experiment 2) to be a major factor that influenced hepatic TG concentration during our experiments. Additionally, we do not know of any evidence that lipolysis of intracellular hepatic TG occurs via a hormone-sensitive lipase similar to that

of adipose tissue. In nonruminants, lipolysis occurs via lysosomal acid lipase; in ruminants, the mechanism and regulation for hepatic lipolysis has not been characterized.

**Effects of hormones.** Estrogen had no effect on hepatocyte uptake of fatty acids from the media or on their partition between oxidation and TG formation (Table 2). The lack of  $E_3$  effects in our experiment could have been due to the absence of responsive estrogen receptors in our cells. However, estrogen receptors have been demonstrated in adult bovine liver (16, 25, 39) and in human fetal hepatocytes (9). The time of exposure to  $E_3$  in our experiment (up to 48 h) should have been sufficient to elicit a response according to Courtney et al. (12) who reported estradiol 17- $\beta$  effects on hepatic rates of lipogenesis and fatty acid esterification as soon as 18 h after in vivo injection. The in vivo effects of estrogen on liver cells possibly were not direct, but were mediated by other hormones or other agents.

Insulin and glucagon had no significant effect on the rates of ASP formation, intracellular TG accumulation, or oleate uptake, but insulin increased ( $P < 0.05$ ) TG accumulation as a proportion of oleate uptake by 22% (main effect of insulin across both oleate concentrations; Table 2). There was no change in the fraction partitioned to ASP ( $P > 0.12$ ). The ratio of oleate converted to ASP and oleate converted to TG was reduced from 2.12 in controls to 1.42 in cells treated with insulin ( $P < 0.03$ ).

In monogastric animals, insulin reduces fatty acid oxidation in the liver by elevating cellular concentrations of malonyl-coenzyme A, a known inhibitor of carnitine acyltransferase. Zammit (43) suggested that insulin could have the same effects in ruminants, but our results do not support that suggestion. In rats and humans, insulin increases hepatic TG synthesis but reduces the rate of secretion of very low density lipoprotein TG (15, 20). Our data agreed with previous reports (18, 33) that insulin has no effect on rates of TG accumulation.

Extracellular fatty acid concentration might be an important regulator of insulin action in the liver (38). Insulin binding and degradation, but not apparent receptor affinity, were reduced when cells were incubated for 45 min in the presence of 0.4 mM oleate (38). Witters and Trasko (42) noted that the insulin effects on fatty acid metabolism were restricted to NEFA concentrations  $<1$  mM, and those researchers could only detect the effects of this hormone when glucagon was also present in the medium. In our experiment, insulin increased the proportion of oleate that was esterified to intracellular TG at extracellular concentrations of 1 and 2 mM ( $P < 0.03$ ), but we did

TABLE 2. Influence of oleate concentrations in media and estriol, insulin, or glucagon on the rates of oleate uptake and conversion to media acid-soluble products (ASP) or intracellular triglyceride (TG) (Experiment 1).

| Oleate concentration<br>and hormone<br>treatment | Uptake <sup>1,a</sup> |      | ASP <sup>1,a</sup> |      | ASP <sup>2</sup> |     | TG <sup>1,a</sup> |      | TG <sup>2,a,b</sup> |     | ASP:TG <sup>3,a,b</sup> |      |
|--------------------------------------------------|-----------------------|------|--------------------|------|------------------|-----|-------------------|------|---------------------|-----|-------------------------|------|
|                                                  | $\bar{X}$             | SE   | $\bar{X}$          | SE   | $\bar{X}$        | SE  | $\bar{X}$         | SE   | $\bar{X}$           | SE  | $\bar{X}$               | SE   |
| 1.0 mM Oleate                                    |                       |      |                    |      |                  |     |                   |      |                     |     |                         |      |
| Control                                          | 1.23                  | 0.25 | 0.72               | 0.16 | 57.8             | 2.3 | 0.24              | 0.06 | 20.4                | 3.3 | 3.06                    | 0.69 |
| Estriol (1 nM)                                   | 1.31                  | 0.21 | 0.72               | 0.14 | 53.9             | 2.9 | 0.26              | 0.06 | 20.1                | 2.8 | 2.80                    | 0.48 |
| Estriol (10 nM)                                  | 1.24                  | 0.29 | 0.70               | 0.16 | 56.3             | 0.9 | 0.26              | 0.06 | 21.6                | 2.8 | 2.71                    | 0.43 |
| Insulin (10 nM)                                  | 1.15                  | 0.20 | 0.61               | 0.14 | 52.0             | 6.6 | 0.34              | 0.04 | 30.9                | 3.1 | 1.75                    | 0.35 |
| Glucagon (1 nM)                                  | 1.37                  | 0.22 | 0.70               | 0.10 | 51.2             | 1.7 | 0.21              | 0.01 | 16.4                | 2.8 | 3.28                    | 0.44 |
| Mean                                             | 1.26                  | 0.09 | 0.69               | 0.06 | 54.2             | 1.2 | 0.26              | 0.02 | 21.9                | 1.7 | 2.72                    | 0.23 |
| 2.0 mM Oleate                                    |                       |      |                    |      |                  |     |                   |      |                     |     |                         |      |
| Control                                          | 2.30                  | 0.58 | 1.18               | 0.37 | 48.3             | 4.6 | 0.94              | 0.17 | 42.5                | 4.8 | 1.18                    | 0.26 |
| Estriol (1 nM)                                   | 2.20                  | 0.44 | 1.27               | 0.28 | 57.7             | 8.5 | 0.76              | 0.15 | 34.7                | 2.4 | 1.65                    | 0.12 |
| Estriol (10 nM)                                  | 2.10                  | 0.51 | 1.05               | 0.31 | 48.3             | 6.6 | 0.85              | 0.10 | 44.0                | 7.7 | 1.18                    | 0.27 |
| Insulin (10 nM)                                  | 2.30                  | 0.51 | 1.17               | 0.32 | 47.1             | 5.2 | 1.03              | 0.12 | 45.9                | 6.6 | 1.09                    | 0.24 |
| Glucagon (1 nM)                                  | 2.40                  | 0.51 | 1.31               | 0.31 | 50.5             | 2.4 | 0.94              | 0.14 | 31.7                | 4.2 | 1.41                    | 0.19 |
| Mean                                             | 2.32                  | 0.20 | 1.20               | 0.12 | 50.4             | 2.4 | 0.90              | 0.06 | 40.8                | 2.4 | 1.30                    | 0.10 |

<sup>a</sup>Main effect of oleate concentration was significant ( $P < 0.001$ ).

<sup>b</sup>Main effect of insulin (vs. control) for TG percentage and the ratio of ASP:TG was significant ( $P < 0.03$ ).

<sup>1</sup>Rates of oleate uptake and conversion to ASP and intracellular TG are means ( $\pm$ SE) of three observations in nanomoles of oleate per microgram of DNA per hour; oleate uptake was calculated as the sum of ASP formation and total intracellular lipid accumulation.

<sup>2</sup>Proportion of oleate partitioned to ASP or to intracellular TG as a percentage of the total oleate uptake.

<sup>3</sup>Ratio between the rates of oleate converted to ASP and intracellular TG.

not observe a significant interaction of insulin and oleate.

Glucagon enhancement of ketogenesis has been reported consistently in rats (5, 42) and humans (41); its negative effect on esterification was noted by Witters and Trasko (42) and by Vons et al. (41), but not by Björnsson et al. (5). Hepatocytes were responsive to glucagon [Experiment 3; (14)]. However, in agreement with the results of Pogson et al. (34), we did not detect an effect of glucagon on fatty acid metabolism.

## Experiment 2: Disposal of ICL

At the end of the loading period, there were no differences in the amount of DNA per plate ( $40.1 \pm 4.0$  and  $41.5 \pm 3.8 \mu\text{g}$ ) or oleate converted to ICL ( $12.1 \pm 1.2$  and  $12.4 \pm 1.2 \text{ nmol}$ ) in cells loaded with and without 10 nM  $\text{E}_3$ , respectively. During the 24 h allowed for depletion of ICL, the mean reduction in DNA content was 16.8% per plate, which was unrelated to conditions during loading, presence of oleate during depletion, or hormonal treatments applied (Table 3) but was likely related to cell senescence and detachment. Treatment with  $\text{E}_3$  during loading had no effect on the amount of ICL remaining in the cells or on the amount and proportion of ICL oxidized to ASP during the depletion period.

**Effects of extracellular NEFA during depletion of intracellular TG.** The presence of oleate during the depletion period increased ( $P < 0.01$ ) the mass of previously accumulated ICL that remained in the cells (Table 3). In the media that were supplemented with oleate, 80.5% of the labeled ICL remained in the plates after the 24 h allowed for depletion; only 70.8% were recovered from the plates when depletion occurred in the absence of oleate ( $P < 0.01$ ; Table 3). The formation of ASP from ICL, expressed either as nanomoles per microgram of DNA or as a percentage of available ICL per plate, was larger when no oleate was present during depletion than when 2 mM oleate was present ( $1.49$  vs.  $1.20 \text{ nmol}/\mu\text{g}$  of DNA;  $P < 0.01$  and  $9.6\%$  vs.  $7.7\%$ ,  $P < 0.05$ ) (Table 3).

The recoveries of label in ASP and ICL after depletion (Table 3) were calculated on a per plate basis, and, in most instances, their sum was  $<90\%$ . On average,  $25.8 \pm 4.1\%$  of the previously accumulated intracellular label appeared in the media at the end of the depletion period, and  $51.3 \pm 2.7\%$  of this could be accounted for by cell lysis (estimated by the disappearance of DNA). Oleate depleted as ASP represented only 36.2% of the previously accumulated intracellular label released to the media. Therefore, part of the ASP in the media could have originated from TG released to the media after cell lysis followed

by extracellular hydrolysis, cellular uptake, and oxidation of FFA by the cells, rather than from intracellular hydrolysis of TG followed by  $\beta$ -oxidation. Therefore, the ASP values reported are the maximum estimate of intracellular TG disposal through oxidation of ICL.

Because ASP in the depletion media could account for <40% of the total label present, we decided to determine whether lipids had been exported from the cells as lipoproteins. After fractionation by gel filtration chromatography, we could not detect any lipoproteins present in the depletion media. Based on the previous discussion, we assume that the secretion of lipoprotein was negligible in our cultures, there was no substantial uptake of TG from the media after cell lysis, and the ASP appearing in the media were originated by oxidation of intracellular TG. The rates of TG oxidation to ASP were less than the rates of oleate conversion to ASP measured between 24 and 48 h in the same cell preparations incubated with 0.1 mM oleate (2.29 nmol of oleate/ $\mu$ g of DNA per 24 h;

Experiment 1). The total label present in the media at the end of the 24 h of depletion was equivalent to 0.12  $\mu$ mol of oleate per plate or 0.08 mM. This amount represents only 4% of the total media NEFA when 2 mM oleate was added during the depletion.

As previously noted, the rates of oleate conversion to ASP remained proportional to the substrate concentration up to 2 mM; therefore, the reduction of ASP formation as oleate concentration increased was probably not attributable to inhibition of  $\beta$ -oxidation by high oleate concentration but to direct inhibition of TG hydrolysis by free oleate. Our findings are in agreement with those of Butler et al. (7), who reported a significant decrease in the production of ketones from endogenous fatty acids by caudate lobes from sheep liver when palmitate in perfusate was increased from 0 to 1.2 mM.

**Effects of insulin, glucagon, and estrogens during depletion of intracellular TG.** No effects of glucagon or E<sub>3</sub> during depletion were observed. Insulin reduced the mass and the proportion of ICL oxi-

TABLE 3. Influence of estriol, insulin, and glucagon on the depletion of prelabeled intracellular lipids (ICL) and conversion of prelabeled ICL to acid-soluble products (ASP) (Experiment 2).

| Oleate concentration<br>and hormone<br>treatment <sup>1</sup> | DNA                  |      | ASP <sup>2,a,b</sup>   |      | ASP <sup>3,a,b</sup> |     | ICL <sup>2,a,c</sup>   |     | ICL <sup>4,b</sup> |      |
|---------------------------------------------------------------|----------------------|------|------------------------|------|----------------------|-----|------------------------|-----|--------------------|------|
|                                                               | ( $\mu$ g per plate) |      | (nmol/ $\mu$ g of DNA) |      | (%)                  |     | (nmol/ $\mu$ g of DNA) |     | (%)                |      |
|                                                               | $\bar{X}$            | SE   | $\bar{X}$              | SE   | $\bar{X}$            | SE  | $\bar{X}$              | SE  | $\bar{X}$          | SE   |
| 1.0 mM Oleate                                                 |                      |      |                        |      |                      |     |                        |     |                    |      |
| Control                                                       | 33.2                 | 4.1  | 1.70                   | 0.37 | 10.8                 | 1.3 | 11.3                   | 3.2 | 66.8               | 1.2  |
| Estriol (1 nM)                                                | 36.0                 | 2.2  | 1.55                   | 0.38 | 10.7                 | 1.6 | 11.8                   | 3.8 | 73.9               | 3.2  |
| Estriol (10 nM)                                               | 32.7                 | 6.5  | 1.53                   | 0.36 | 8.7                  | 1.4 | 12.0                   | 2.9 | 65.0               | 5.5  |
| Insulin (10 nM)                                               | 31.8                 | 9.1  | 0.75                   | 0.33 | 6.2                  | 4.4 | 14.2                   | 7.0 | 81.4               | 9.6  |
| Glucagon (1 nM)                                               | 35.6                 | 9.4  | 1.64                   | 0.85 | 10.4                 | 1.4 | 11.3                   | 5.3 | 73.4               | 4.7  |
| Mean                                                          | 33.9                 | 2.2  | 1.49                   | 0.18 | 9.6                  | 0.8 | 12.0                   | 1.5 | 70.8               | 2.2  |
| 2.0 mM Oleate                                                 |                      |      |                        |      |                      |     |                        |     |                    |      |
| Control                                                       | 39.5                 | 6.2  | 1.26                   | 0.33 | 9.0                  | 0.6 | 11.9                   | 3.1 | 84.3               | 2.9  |
| Estriol (1 nM)                                                | 36.5                 | 5.0  | 1.18                   | 0.31 | 8.0                  | 1.0 | 12.7                   | 3.7 | 81.5               | 3.6  |
| Estriol (10 nM)                                               | 31.3                 | 4.5  | 1.36                   | 0.43 | 7.6                  | 0.9 | 15.0                   | 5.0 | 79.6               | 9.4  |
| Insulin (10 nM)                                               | 31.9                 | 10.9 | 0.68                   | 0.27 | 4.1                  | 0.3 | 14.0                   | 7.0 | 79.3               | 14.9 |
| Glucagon (1 nM)                                               | 32.0                 | 7.3  | 1.30                   | 0.38 | 8.6                  | 1.8 | 13.5                   | 7.3 | 74.3               | 9.5  |
| Mean                                                          | 34.8                 | 2.5  | 1.20                   | 0.16 | 7.7                  | 0.5 | 13.3                   | 1.8 | 80.5               | 2.9  |

<sup>a</sup>Main effect of insulin versus control was significant ( $P < 0.01$ ).

<sup>b</sup>Main effect of oleate was significant ( $P < 0.01$ ).

<sup>c</sup>Main effect of oleate was significant ( $P < 0.05$ ).

<sup>1</sup>Plated calf hepatocytes cultured in Dulbecco's modified Eagle's medium supplemented with 2 mM [1-<sup>14</sup>C]oleate for 24 h were allowed to mobilize stored ICL during the following 24 h in the presence of 0 or 2 mM oleate and estriol, insulin, or glucagon or no hormone.

<sup>2</sup>Values are means ( $\pm$ SE) of two observations in nanomoles of ICL oleate oxidized to ASP per microgram of DNA per 24 h or nanomoles of ICL oleate remaining intracellularly per microgram of DNA.

<sup>3</sup>Percentage of the lipids present intracellularly at the end of the 24-h loading period that was depleted as ASP after the depletion period from 24 to 48 h; calculated as nanomoles of ASP per plate at the end of depletion  $\times$  100/nanomoles of ICL oleate per plate at the end of loading.

<sup>4</sup>Percentage of lipids present intracellularly at the end of the 24-h loading period that remained intracellularly after the depletion period from 24 to 48 h; calculated as nanomoles of ICL oleate per plate at the end of depletion  $\times$  100/nanomoles of ICL oleate per plate at the end of loading.

dized to ASP by 49% (from 1.48 to 0.72 nmol of oleate converted to ASP/ $\mu$ g of DNA per 24 h and from 9.9 to 5.2% of the available ICL per plate;  $P < 0.01$ ; Table 3). Insulin also reduced the total label released to the media; the ICL remaining intracellularly increased from 11.6 nmol/ $\mu$ g of DNA in controls to 14.1 nmol/ $\mu$ g of DNA in plates treated with insulin ( $P < 0.1$ ; Table 3).

Duerden and Gibbons (15) applied the same experimental strategy used in our depletion experiments to evaluate the effect of insulin on the transfer of intracytoplasmic TG to very low density lipoprotein TG by rat hepatocytes. From 24 to 48 h, control cells were able to export 45.4% of the available TG as very low density lipoprotein TG, and cells incubated with insulin (78 nM, compared with 10 nM used in our experiment) exported 20.4%. The incorporation of intracytoplasmic TG to very low density lipoproteins involves the hydrolysis of TG and its resynthesis at the endoplasmic reticulum. Insulin might inhibit intracytoplasmic TG export by acting on lysosomal acid lipase, which would be consistent with the 49% reduction in the rate of TG disposal through oxidation to ASP in our experiment. We did not detect significant

inhibition of ketogenesis by insulin in Experiment 1; therefore, the observed impairment in the disposal of ICL through oxidation might have been due to an inhibition of TG hydrolysis induced by insulin.

### Experiment 3: Ketogenesis and Gluconeogenesis in Cells Infiltrated with Lipid

Cells incubated with 2 mM oleate for 45 h incorporated a mean of 37.9  $\mu$ g of oleate to intracellular TG per microgram of DNA, which was equivalent to 12.5% (wet basis), assuming the ratio of protein:DNA was 45 (13) and that cells contained 15% protein [wet basis; (7)]. Such culture conditions resulted in excess cytoplasmic TG compared with cells that were incubated without oleate. Therefore, these cells will be referred to as cells infiltrated with lipid.

**Gluconeogenesis.** In cells infiltrated with lipid, the mean rate of propionate conversion to media glucose was 1.46 nmol of glucose/ $\mu$ g of DNA per h, which was 24% lower ( $P < 0.05$ ) than the mean rate in control cells incubated with no oleate from 0 to 45 h (1.92 nmol of glucose/ $\mu$ g of DNA per h; Table 4). The

TABLE 4. Effect of intracellular lipid (ICL) infiltration by preexposure to oleate and concurrent presence of oleate on the rate of propionate conversion to glucose and ICL (Experiment 3).<sup>1</sup>

|                                        | Oleate (45 to 48 h)  |       |           |       |                         |       |           |       |
|----------------------------------------|----------------------|-------|-----------|-------|-------------------------|-------|-----------|-------|
|                                        | Noninfiltrated cells |       |           |       | Lipid infiltrated cells |       |           |       |
|                                        | 0 mM                 |       | 2 mM      |       | 0 mM                    |       | 2 mM      |       |
|                                        | $\bar{X}$            | SE    | $\bar{X}$ | SE    | $\bar{X}$               | SE    | $\bar{X}$ | SE    |
| Propionate conversion to media glucose |                      |       |           |       |                         |       |           |       |
| Control                                | 1.68                 | 0.82  | 1.83      | 0.98  | 1.20                    | 0.82  | 1.17      | 0.43  |
| Insulin (10 nM)                        | 2.13                 | 0.98  | 1.35      | 0.54  | 1.71                    | 0.98  | 0.66      | 0.03  |
| Glucagon (1 nM) <sup>a</sup>           | 2.44                 | 1.06  | 2.10      | 1.13  | 2.30                    | 1.06  | 1.71      | 0.79  |
| Mean <sup>b,c</sup>                    | 2.08                 | 0.45  | 1.76      | 0.43  | 1.74                    | 0.39  | 1.18      | 0.30  |
| Propionate conversion to ICL           |                      |       |           |       |                         |       |           |       |
| Control                                | 0.031                | 0.001 | 0.114     | 0.043 | 0.072                   | 0.027 | 0.137     | 0.069 |
| Insulin (10 nM)                        | 0.030                | 0.006 | 0.101     | 0.008 | 0.048                   | 0.008 | 0.114     | 0.008 |
| Glucagon (1 nM)                        | 0.025                | 0.007 | 0.093     | 0.023 | 0.065                   | 0.023 | 0.125     | 0.048 |
| Mean <sup>b,d</sup>                    | 0.029                | 0.004 | 0.102     | 0.019 | 0.062                   | 0.010 | 0.125     | 0.022 |
| Total propionate incorporation         |                      |       |           |       |                         |       |           |       |
| Control                                | 1.71                 | 0.83  | 1.94      | 1.03  | 1.28                    | 0.49  | 1.31      | 0.49  |
| Insulin (10 nM)                        | 2.15                 | 0.99  | 1.45      | 0.58  | 1.78                    | 0.69  | 0.78      | 0.08  |
| Glucagon (1 nM) <sup>a</sup>           | 2.47                 | 1.06  | 2.19      | 1.17  | 2.35                    | 1.01  | 1.83      | 0.80  |
| Mean <sup>b</sup>                      | 2.11                 | 0.45  | 1.86      | 0.45  | 1.80                    | 0.39  | 1.31      | 0.31  |

<sup>a</sup>Cells treated with glucagon were significantly different from control cells with no hormone ( $P < 0.005$ ).

<sup>b</sup>Main effect of preincubation with oleate was significant ( $P < 0.005$ ).

<sup>c</sup>Main effect of concurrent oleate was significant ( $P < 0.005$ ).

<sup>d</sup>Main effect of concurrent oleate was significant ( $P < 0.001$ ).

<sup>1</sup>Rates of propionate conversion to glucose or ICL (nanomoles of [<sup>14</sup>C]propionate converted per microgram of DNA per hour) were measured between 45 and 48 h in the presence of 0 or 2 mM oleate and several hormone additions in plated calf hepatocytes cultured in Dulbecco's modified Eagle's medium supplemented with or without 2 mM oleate from 0 to 45 h. Values are means ( $\pm$ SE) of two observations.



TABLE 5. Effect of preexposure to oleate and concurrent presence of propionate on the rate of oleate conversion to media ASP (Experiment 3).<sup>1</sup>

|                     | Propionate from 45 to 48 h |      |           |      |                         |      |           |      |
|---------------------|----------------------------|------|-----------|------|-------------------------|------|-----------|------|
|                     | Noninfiltrated cells       |      |           |      | Lipid infiltrated cells |      |           |      |
|                     | 0                          |      | 2.5 mM    |      | 0                       |      | 2.5 mM    |      |
|                     | $\bar{X}$                  | SE   | $\bar{X}$ | SE   | $\bar{X}$               | SE   | $\bar{X}$ | SE   |
| Control             | 1.08                       | 0.01 | 0.63      | 0.17 | 1.49                    | 0.33 | 1.09      | 0.04 |
| Insulin (10 nM)     | 1.15                       | 0.10 | 0.71      | 0.06 | 1.49                    | 0.00 | 1.06      | 0.22 |
| Glucagon (1 nM)     | 1.19                       | 0.07 | 0.63      | 0.06 | 1.71                    | 0.14 | 1.07      | 0.19 |
| Mean <sup>a,b</sup> | 1.14                       | 0.04 | 0.66      | 0.03 | 1.57                    | 0.11 | 1.07      | 0.08 |

<sup>a</sup>Main effect of lipid infiltration was significant ( $P < 0.001$ ).

<sup>b</sup>Main effect of propionate was significant ( $P < 0.001$ ).

<sup>1</sup>Rates of ASP formation from media oleate (nanomoles of [2-<sup>14</sup>C]oleate converted to ASP per microgram of DNA per hour) were measured between 45 and 48 h in the presence of 0 or 2.5 mM propionate and several hormone additions in plated calf hepatocytes cultured in Dulbecco's modified Eagle's medium supplemented with or without 2 mM oleate from 0 to 45 h. Values are means ( $\pm$ SE) of two observations.

concurrent presence of 2 mM oleate during the 3-h period for which gluconeogenesis was measured reduced the rate of propionate conversion to media glucose by 23% ( $P < 0.05$ ), from 1.91 to 1.47 nmol of glucose/ $\mu$ g of DNA per h. The effects of fat infiltration and of concurrent oleate presence were of the same magnitude and were additive and independent of each other.

To determine whether gluconeogenic carbon was being diverted for fatty acid esterification at the glycerol phosphate step, we measured propionate incorporation to ICL. More propionate was transferred to glycerol when oleate was concurrently present ( $P < 0.001$ ) and in cells infiltrated with lipid ( $P < 0.05$ ; Table 4) without significant interaction between the two factors. On average, in uninfiltrated cells incubated without oleate between 45 and 48 h, 1.4% of the total propionate incorporated into glucose or ICL was in ICL. When the cells were preincubated with oleate and allowed to accumulate ICL, an extra 2% of propionate appeared in ICL, but when oleate was present between 45 and 48 h, 5.8% of the propionate appeared in ICL. These observations indicate that some recycling of intracellular TG occurred and that fatty acid esterification could represent a reduction of gluconeogenic carbon. The lack of interaction between preincubation conditions and oleate presence in the media suggests that hydrolysis of intracellular TG was not inhibited by high extracellular concentrations of fatty acids, which is in contrast to the observed reduction in ASP formation from intracellular TG in Experiment 2. However, the 3 h allowed for recycling of ICL in the present experiment (Experiment 3) might have been too short to permit detection of the proposed inhibitory effect.

The reduction in the rate of gluconeogenesis (glucose production in media) when oleate was present is contradictory with a previous report (1) that rates of gluconeogenesis in goats increased linearly with oleate concentrations up to 2 mM. In that experiment, short-term suspensions were used, and cells were incubated in Krebs buffer supplemented with propionate and oleate; the latter was the main source of reducing equivalents for gluconeogenesis. In the present experiment, pyruvate and lactate in the media could have satisfied the requirement for reducing equivalents, independently of oleate. Using a culture system similar to ours, Strang et al. (37) did not observe an effect of fatty acids on gluconeogenesis. Those researchers used a physiological mixture of fatty acids instead of oleate; therefore, fatty acid profile might influence propionate conversion to glucose.

Slices from livers that accumulated TG in vivo exhibited decreased gluconeogenic capacity (32), which could have been due to fat infiltration but also to the conditions that led to fatty liver. Our results suggest that fat infiltration has a direct effect on and at least partially accounts for a reduction in liver gluconeogenic capacity. Because oxidation of intracytoplasmic TG is very slow in bovine hepatocytes (Experiment 2), the negative effect of preexposure to fatty acids on gluconeogenesis is probably not related to products that arise by oxidation of intracellular TG.

The rate of gluconeogenesis was increased by glucagon ( $2.14 \pm 0.39$  vs.  $1.47 \pm 0.29$  nmol [1-<sup>14</sup>C]propionate converted to glucose per microgram of DNA per hour;  $P < 0.02$ ), which agrees with the results of a study by Donkin and Armentano (14). This effect was independent of preincubation condi-

tions or presence of oleate. In rats, oleate in the media masked glucagon stimulation of gluconeogenesis from [2-<sup>14</sup>C]propionate (6). Interactions among preexposure to fatty acids, oleate presence, or hormonal additions were not detected.

No response to insulin was observed, which disagrees with the results of a study by Donkin and Armentano (14). Cells for this experiment were prepared from different calves than those used for Experiment 2, which proved responsive to insulin. Insulin responsiveness might have been lost in the cell preparations; retention of this cell function could be inconsistent among cell preparations (34).

**ASP formation.** The rate of ASP formation was lower in cells incubated in the presence of propionate (0.86 vs. 1.35 nmol of oleate converted to ASP per microgram of DNA per hour;  $P < 0.001$ ; Table 5) and in those that had not been preincubated with 2 mM oleate for the preceding 45 h (0.90 vs. 1.32 nmol oleate converted to ASP per microgram of DNA per hour;  $P < 0.001$ ). The negative effect of propionate on ketone production by liver cells of the ruminant (2, 30) and nonruminant (27) is well documented. Preincubation with 2 mM oleate increased the rate of ASP formation by 48%. Oleate metabolism by hepatocytes from goats in negative and positive energy balance were similar despite the fact that plasma NEFA were increased fourfold and liver TG were increased twofold in goats fed restricted amounts of energy at the time of cell isolation (2). In the present experiments, the rate of ASP formation from 24 to 48 h did not change for any oleate concentration (Experiment 1; Table 1) despite considerable TG accumulation during this period. The net production of ketone bodies across the liver increased from 2.15 mmol/min in fed cows to 4.33 mmol/min in lactating cows that had been deprived of feed for 6 d (3). When perfused with 2 mM oleate, livers from rats fed diets containing 90% fat (wt/wt) had rates of ketogenesis that were 30% higher than those of livers from starved rats, which was attributed to increased capacity of hydroxymethylglutaryl-coenzyme synthase (27). Therefore, exposure to extracellular NEFA, rather than the intracellular fat accumulation, might be responsible for the increase in the rate of ASP formation. In Experiment 2, the presence of 2 mM oleate during the depletion phase reduced the oxidation of ICL to media ASP by 25%. In contrast, in Experiment 3, preexposure to 2 mM oleate and TG accumulation resulted in increased fatty acid oxidation. This result reinforces our conclusion that the presence of oleate during depletion impaired TG disposal by inhibiting TG hydrolysis rather than by decreasing the rate of  $\beta$ -oxidation.

Propionate did not interact with preexposure to oleate on the formation of ASP (Table 5). The hormonal additions assayed had no significant effects on the rate of ASP formation.

## CONCLUSIONS

Although the rate of fatty acid uptake by hepatocytes remained constant as extracellular concentration of NEFA increased, larger proportions were partitioned toward esterification. Glucagon and  $E_3$  did not affect the rate of uptake or the relative partition of fatty acids within the hepatocyte, but insulin increased the proportion esterified. The disposal of ICL was very slow compared with the rates of extracellular fatty acid utilization by bovine hepatocytes. The disposal of ICL was accomplished through oxidation and was inhibited by the presence of extracellular fatty acids or by insulin. The exposure to high concentrations of NEFA, with subsequent accumulation of intracellular TG, reduced the rate of gluconeogenesis and enhanced formation of ASP.

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